

Department of Control Science and Engineering
University of Science and Technology Beijing
Beijing 100083, China

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Editor-in-Chief
Computers & Industrial Engineering

Dear Editor-in-Chief,

Please consider our Research Paper entitled “Two-stage MILP scheduling and structure-aware learning-enhanced branch-and-cut for dual-source mixed-flow rotating-chamber cluster tools” for publication in *Computers & Industrial Engineering*.

The paper addresses an industrial scheduling problem arising in rotating-chamber cluster tools used in a PECVD process setting. Product wafers and reusable vacuum-zone dummy wafers share chambers, load locks, and robots while obeying recipe grouping, residency, cleaning, capacity, and transfer constraints. The resulting dual-source mixed-flow problem is not represented by existing single-flow or fixed-routing cluster-tool models.

The paper makes two connected contributions. First, it develops a finite-horizon two-stage MILP that jointly determines recipe grouping, chamber and slot assignment, transfers, cleaning, dummy-wafer allocation, and timing; the second stage preserves the stage-one makespan within numerical tolerance and the discrete schedule while refining physical route waits and processing cadence. Second, it develops RDMCT-HBS, a structure-aware learning-enhanced branch-and-cut method combining an SPBS warm start, reconstructed variable-role features, A3C-inspired parallel policy-gradient training, bounded beam decoding, and role-submodular completion.

The computational evidence is both extensive and mechanism-specific. In a 2700-run nominal comparison, RDMCT-HBS records the lowest mean primal–dual integral and mean solution time among seven methods. A separate 960-run controlled experiment shows that SPBS reduces mean primal–dual integral by 26.19% and increases incumbent availability from 51.67% to 100%. In 1500 matched stability runs, two-stage refinement reduces physical route-total wait, route-maximum wait, cadence coefficient of variation, and cadence deviation by 51.91%, 68.89%, 39.70%, and 94.05%, respectively, relative to disabling refinement. A complete 300-run factorial experiment further identifies the interaction between wafer count and mixed-flow recipe as the principal source of computational difficulty in the factorial design. The larger-scale stress test adds 729 time-limited runs on nine 40–64-wafer configurations, every one of which returns an independently validated incumbent. Across the complete evidence programme, all 6273 planned runs are accounted for and all 6041 saved incumbents pass independent feasibility checking.

The work fits the journal because it integrates a new industrial-engineering model with an explainable, exact-optimization-compatible learning method and evaluates both computational and manufacturing consequences. The formulation and solution architecture are also transferable to other tightly coupled multi-resource scheduling systems with auxiliary flows and shared transport.

Thank you for your consideration.

Sincerely,

Lizhen Shao
Corresponding author
University of Science and Technology
Beijing
lshao@ustb.edu.cn